

Beaufort Keystream Anomalies in Kryptos K4

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Abstract

We report model-independent statistical anomalies in the Beaufort keystream of Kryptos K4 at its 24 known crib positions. The keystream values $k[i] = (C[i] + P[i]) \bmod 26$ are enriched for the 7-letter set $\Pi = \{B, G, I, K, O, W, Z\}$: 13/24 overall ($p = 4.3 \times 10^{-3}$), with extreme concentration at BERLINCLOCK positions 63–70 (7/8, $p = 6.3 \times 10^{-4}$). These letters correspond to columns 0 and 3 of the Kryptos Alphabet arranged as a 5-wide Polybius grid. Bean (2021) independently found that 13/24 keystream values are multiples of 5 in a reversed-KA indexing ($p \approx 9.5 \times 10^{-4}$), sharing a mod-5 connection with the grid. All keystream findings depend only on the carved ciphertext and the known cribs—no cipher model, keyword, or null mask is assumed. A previously reported null-palette restriction (7 distinct letters at 17 consensus null positions) has been **downgraded**: cross-model testing shows the restriction is an artifact of the iterative optimization procedure, not a model-invariant property of K4 (Section 5).

1 Setup

Let $\mathcal{A} = \{A, B, \dots, Z\}$ with standard indexing $A=0, \dots, Z=25$. All arithmetic modulo 26.

Definition 1 (Kryptos Alphabet). $KA = KRYPTOSABCDEFGHIJLMNQUVWXZ$, with $KA[0]=K$, $KA[1]=R$, $\dots$, $KA[25]=Z$. All 26 letters appear exactly once.

Definition 2 (K4 Ciphertext and Cribs).

$C = OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSQSSEKZZWAT$
 $JKLUDIAWINFBNYPVTMTZFPKWGDKZXTJCDIGKUHUAUEKCAR$

with $|C| = 97$ and 0-indexed positions. Known cribs (24 characters total):

$P[21..33] = EASTNORTHEAST \quad (13 \text{ chars}),$

$P[63..73] = BERLINCLOCK \quad (11 \text{ chars}).$

Definition 3 (5-Wide KA Polybius Grid). Arrange KA into a grid with 5 columns:

	c_0	c_1	c_2	c_3	c_4
r_0	K	R	Y	P	T
r_1	O	S	A	B	C
r_2	D	E	F	G	H
r_3	I	J	L	M	N
r_4	Q	U	V	W	X
r_5	Z				

where bold letters occupy columns 0 and 3 exclusively. For letter ℓ , define $\text{col}(\ell) = \text{KA-idx}(\ell) \bmod 5$.

Definition 4 (Palette). $\Pi = \{\ell \in \mathcal{A} : \text{col}(\ell) \in \{0, 3\}\} = \{B, G, I, K, O, W, Z\}$.

2 Proposition 1: Beaufort Keystream Enrichment

Definition 5 (Beaufort Keystream at Crib Positions). *For each known crib position i :*

$$k[i] = (C[i] + P[i]) \bmod 26.$$

*This value is **model-independent**: it depends only on the carved text and the known plaintext, not on any cipher model, keyword, or null mask.*

Proposition 1 (Palette Enrichment). *Of the 24 Beaufort keystream values at crib positions, 13 are members of Π :*

$$|\{i \in \mathcal{C} : k[i] \in \Pi\}| = 13.$$

Proof. Direct computation at all 24 crib positions.

EASTNORTHEAST (positions 21–33):

Pos	$C[i]$	$P[i]$	$k[i]$	$\in \Pi?$
21	F	E	J (9)	No
22	L	A	L (11)	No
23	R	S	J (9)	No
24	V	T	O (14)	Yes
25	Q	N	D (3)	No
26	Q	O	E (4)	No
27	P	R	G (6)	Yes
28	R	T	K (10)	Yes
29	N	H	U (20)	No
30	G	E	K (10)	Yes
31	K	A	K (10)	Yes
32	S	S	K (10)	Yes
33	S	T	L (11)	No

ENE palette hits: 6/13.

BERLINCLOCK (positions 63–73):

Pos	$C[i]$	$P[i]$	$k[i]$	$\in \Pi?$
63	N	B	O (14)	Yes
64	Y	E	C (2)	No
65	P	R	G (6)	Yes
66	V	L	G (6)	Yes
67	T	I	B (1)	Yes
68	T	N	G (6)	Yes
69	M	C	O (14)	Yes
70	Z	L	K (10)	Yes
71	F	O	T (19)	No
72	P	C	R (17)	No
73	K	K	U (20)	No

BCL palette hits: 7/11. Total: 13/24.

□

Statistical test. Under H_0 (keystream uniform over $\mathcal{A}$), $P(\text{palette letter}) = |\Pi|/26 = 7/26 \approx 0.269$.

$$P(X \geq 13 \mid n=24, p=7/26) = 4.3 \times 10^{-3}.$$

For the BCL first 8 positions specifically:

$$P(X \geq 7 \mid n=8, p=7/26) = 6.3 \times 10^{-4}.$$

Variant specificity. Only Beaufort ($A=0$) achieves 13/24. Vigenère gives 9/24; Variant Beaufort gives 5/24.

Search burden. Three cipher variants were tested. After Bonferroni correction ($\times 3$), the BCL first-8 result gives $p = 1.9 \times 10^{-3}$, still significant. The palette Π was identified from the null-position analysis (Section 5), making the keystream test partially pre-specified (the palette was defined before the keystream was examined, on disjoint positions) but ultimately derived from the same ciphertext.

3 Bean’s Mod-5 Observation (2021)

Proposition 2 (Bean’s Mod-5 Observation). *Define the reversed Kryptos Alphabet $\text{rKA} = \text{ZXWVUQNMLJIHGFEDCBASOTPYRK}$ and compute:*

$$\beta[i] = (\text{rKA-idx}(C[i]) - \text{idx}(P[i])) \bmod 26$$

at each crib position i . Then 13 of 24 values satisfy $\beta[i] \equiv 0 \pmod{5}$.

Proof. Bean (2021, footnote 3). We independently verify: $P(X \geq 13 \mid n=24, p=6/26) = 9.5 \times 10^{-4}$. $\square$

Apparent structural connection (incorrect). An earlier draft of this paper claimed that $\beta[i] \equiv 0 \pmod{5}$ is equivalent to $\text{col}(C[i]) \equiv -\text{idx}(P[i]) \pmod{5}$, which would directly select KA Polybius columns. This equivalence is **wrong**: it requires that $(x + y) \bmod 26 \equiv (x + y) \bmod 5$, which holds only when $x + y < 26$. Since $26 \equiv 1 \pmod{5}$, reducing mod 26 before mod 5 introduces a ± 1 shift for values ≥ 26 . Empirically, only 9 of the 13 Bean mod-5 hits satisfy the column congruence.

Relationship to the Polybius grid. Bean’s mod-5 and the palette enrichment share a surface connection: both involve divisibility by 5, and the KA Polybius grid has 5 columns. However, the algebraic reduction from Bean’s statistic to column selection does not hold exactly. Bean’s observation is a genuine, independently significant anomaly ($p = 9.5 \times 10^{-4}$) that is *related to* but not *equivalent to* the Polybius column structure.

Independence. Bean’s observation was published in 2021, before the palette Π was identified. The two findings were made by different researchers, from different starting points (reversed-KA keystream arithmetic vs. null-character identity).

4 K-Sum Bias

Proposition 3 (Keystream K-Sum Deficit). *The Beaufort keystream at the 24 crib positions is JLJODEGKUKKKLOCGGBGOKTRU. Mapping each letter to its KA index and summing:*

$$\Sigma = \sum_{i \in \mathcal{C}} \text{KA-idx}(k[i]) = 218.$$

Under a uniform keystream, $E[\Sigma] = 24 \cdot 25/2 = 300$, $\text{SD}(\Sigma) = \sqrt{24 \cdot (26^2 - 1)/12} \approx 36.7$. The deficit is 82, giving $z = -2.23$ ($p \approx 0.013$, one-tailed).

Conditional status. The K-sum deficit is largely a consequence of the palette enrichment (Proposition 1): palette letters Π occupy early KA positions (mean KA-index 7.7 vs. 12.5 overall), so any keystream enriched for Π will mechanically produce a low KA-sum. The K-sum is therefore a *conditional redescription* of the palette signal, not an independent observation.

Note. This observation was first noted by Bean in private correspondence (March 2026), who reported the K-sum as “the only interesting thing about that key.”

5 Downgraded: Null Palette Restriction

An earlier version of this paper presented the null palette restriction as a co-equal finding alongside the keystream enrichment. Cross-model testing has revealed this claim to be **substantially weaker than originally reported**.

Original claim. The 17 consensus null positions (identified by simulated annealing under a KA autokey Vigenère model) contain only 7 distinct letters, all from Π . Monte Carlo testing gave $p \approx 6.3 \times 10^{-5}$.

What went wrong. The 7-distinct result was produced by a two-phase optimization: (1) 200 SA restarts identify high-frequency positions; (2) those positions are *fixed*, and SA optimizes the remaining positions subject to that constraint. Phase 2 mechanically restricts which letters can appear, because fixing early-selected positions removes letter diversity that would otherwise accumulate.

When the full 200-restart protocol is run *without* the bootstrapping phase, even the original discovery model (KA autokey Vigenère) produces 13 distinct letters at its consensus positions—not 7. Across 5 cipher models tested:

Model	Distinct	$ \Pi \cap \text{palette} $
KA autokey Vigenère	13	6/7
KA autokey Beaufort	10	3/7
KA periodic Beaufort	10	6/7
AZ periodic Vigenère	13	4/7
AZ periodic Beaufort	14	4/7

Zero models produce ≤ 7 distinct letters. The mean position Jaccard similarity across models is 0.152, confirming that positions are heavily model-dependent.

What survives. The 0/500 shuffled-CT result remains valid: when K4’s letter arrangement is randomly permuted, SA-optimized masks produce 9–17 distinct letters (mean 12.7), never ≤ 7 . This means real K4 is unusual in *permitting* low-diversity subsets under aggressive optimization—but the optimization procedure itself is responsible for driving diversity below 10.

Implication for the joint significance. The previously reported joint $p = 1.4 \times 10^{-7}$ (combining palette restriction and keystream enrichment) is **no longer valid**, because the palette restriction component has been invalidated. The keystream enrichment (Proposition 1, $p = 4.3 \times 10^{-3}$; BCL first-8: $p = 6.3 \times 10^{-4}$) stands on its own as the primary surviving signal.

6 Summary of Evidence

Observation	Status	p -value	Model-free?
Keystream $13/24 \in \Pi$	Active	4.3×10^{-3}	Yes
BCL first 8: $7/8 \in \Pi$	Active	6.3×10^{-4}	Yes
Bean mod-5 (2021)	Active	9.5×10^{-4}	Yes
K-sum = 218	Conditional	0.013	Yes (but not independent)
Beaufort > other variants	Active	Bonferroni $\times 3$	Yes
Null palette ≤ 7 distinct	Downgraded	was 6.3×10^{-5}	No
Joint $p = 1.4 \times 10^{-7}$	Retracted	—	No

The surviving evidence is moderate: three model-independent signals (keystream palette enrichment, BCL concentration, Bean mod-5) all point toward a mod-5 / KA-column structure in the Beaufort keystream. None individually reaches $p < 10^{-4}$ after search-breadth correction, and none produces a decryption. These are descriptive anomalies that constrain the mechanism but do not identify it.

7 Open Questions

Resolved: Width-21 bigram is a stego artifact. The width-21 bigram signal ($p = 1.6 \times 10^{-4}$) is present in the raw 97-character text but *disappears completely* after removing the 17 consensus null positions.

Resolved: Width-10/17 bigrams are cipher-layer artifacts. On the null-extracted text, bigram enrichment at widths 10 and 17 ($p = 6\text{--}8 \times 10^{-3}$) is destroyed by undoing a column-7 transposition, suggesting mathematical artifact rather than cipher-layer structure.

Open: Why is the Beaufort keystream enriched for Π ? The enrichment is a property of the carved ciphertext at crib positions. Any correct theory of K4 must explain why $(C[i] + P[i]) \bmod 26$ lands in Π at 7/8 BERLINCLOCK positions. Under Beaufort, this means the operational key at those positions is almost entirely drawn from Π .

Open: Running-key source identification. Running key is the only structured non-periodic key model surviving Bean constraints. Keystream fragments at crib positions do not resemble English under identity or columnar transposition (widths 6, 8, 9). Running key with a monoalphabetic inner layer remains underdetermined.

Reproducibility

All computations use the `kryptos` kernel (`src/kryptos/kernel/constants.py`, which self-verifies CT, cribs, and Bean constraints at import time). Cross-model palette stability: `scripts/statistical/e_cross_model_palette`, 5 models $\times$ 200 SA restarts $\times$ 300K steps. Running-key English-likelihood test: `scripts/statistical/e_keystream_eng`, seed 42, $N = 1,000,000$ null fragments, 17,124 columnar orderings. Source: github.com/jcolinpatrick/kryptos. As of March 31, 2026: 940 scripts tracked, 390 eliminations, $671 \times 10^9 +$ configurations tested.

References

- [1] R. Bean, “Cryptodiagnosis of Kryptos K4,” *Proceedings of the 4th International Conference on Historical Cryptology (HistoCrypt 2021)*, pp. 13–21.
- [2] J. Sanborn, Dedication speech, November 3, 1990. RR Auction lot documentation.